

Calculator Trust Test Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	calculator-trust-test

Big idea: Tools need interpretation.

One-Minute Teacher Brief

Students investigate how mathematical technology can both reveal and mislead.

Student Prompt

Inspect Calculator Trust Test Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Calculator Trust Test Stimulus A	Student-facing stimulus 1: A calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Calculator Trust Test Stimulus B	Student-facing stimulus 2: A deliberately mistyped expression that creates a plausible wrong answer. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Calculator Trust Test Reveal / Twist	Student-facing stimulus 3: A trust checklist for tools: input, process, output, and independent check. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Tools need interpretation.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Calculator Trust Test Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Calculator Trust Test Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Does technology make mathematical knowledge more reliable?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how tools need interpretation.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Does technology make mathematical knowledge more reliable?

Setup Checklist

- Prepare a starter stimulus using: A calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- A calculator output checked against estimation, mental maths, and another tool, such as $\log_{10}(1000) = 3$.
- A deliberately mistyped expression that creates a plausible wrong answer.
- A trust checklist for tools: input, process, output, and independent check.

Resources To Use

- Resource pack: <resources/calculator-trust-test/index.html>
- Card deck CSV: <resources/calculator-trust-test/cards.csv>
- Visual prompt: <resources/calculator-trust-test/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Give examples where technology can mislead: restricted graph window, rounded decimals, regression on inappropriate data, or a graph that hides an asymptote.
2. Students explain what the tool displays and what it does not display.
3. They write a warning label for the tool.

4. Discuss responsibility for interpretation.

Debrief Moves

- Knowledge question: Does technology make mathematical knowledge more reliable?
- When can mathematical tools mislead us?
- Who is responsible for interpreting a result: tool or knower?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...